

AstroBin Depth Translator - Technical Documentation

The reasoning, the physics, the formulas, and the implementation.

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Table of contents

1. What this extension is, and the problem it solves
 2. The central idea: depth per unit area of sky
 3. The physics, from photons to a comparable number
 4. The depth formulas
 5. The translator: converting one rig's time to another's
 6. Why focal length and pixel size do not appear
 7. Bands, not filters
 8. Colour sensors and the Bayer matrix
 9. Sky brightness: SQM, Bortle, and the Moon
 10. The reference rig (the universal yardstick)
 11. The headline number: total shooting time
 12. Multi-line filters: dual-band vs. filter sets
 13. The confidence score
 14. A fully worked example
 15. Assumptions and limitations
 16. Where the numbers come from (data sources)
 17. Software architecture
 18. Privacy and analytics
 19. Glossary of symbols
-

1. What this extension is, and the problem it solves

AstroBin is the largest database of astrophotography in the world. Every image lists how long it was exposed - its *integration time* - often as a headline figure like "42 hours." Newcomers and veterans alike use that number as a proxy for "how good" or "how deep" an image is.

It is a deeply misleading proxy.

Integration time only tells you how long the shutter was open. It says nothing about how much light was actually collected, which is what determines how deep an image really goes. Two images of the same target, both at "10 hours," can differ by more than a factor of ten in real signal collected, because the amount of light gathered depends on:

- **Aperture** - a 200 mm telescope collects roughly seven times the light of a 75 mm refractor at the same exposure time.
- **Sky brightness** - an hour under a pristine rural sky is worth many hours from a light-polluted city.
- **Sensor and filter throughput** - quantum efficiency, filter transmission, and filter bandwidth all change how much usable signal lands on the sensor.
- **Whether it is broadband or narrowband**, and how the colour sensor's Bayer matrix splits the light.

So "hours" is the wrong currency. It is like comparing two journeys by how long the engine ran, ignoring the speed of the vehicle.

AstroBin Depth Translator replaces that broken comparison. It adds a small panel to any AstroBin image and answers two questions:

1. **How deep is this image, really?** - expressed as a single, gear-independent number on a fixed reference setup, so any two images become directly comparable at a glance.

2. **How long would it take me to match it?** - the equivalent exposure time on *your own* telescope, camera, filters, and sky, broken down per band.

It is not a judgement of artistic quality, framing, processing, or seeing. It estimates one physical thing: the depth of the underlying data.

2. The central idea: depth per unit area of sky

The quantity that actually matters for deep-sky imaging is **signal-to-noise ratio (SNR)** - how far the faint target signal rises above the random noise in

the image. More integration, more aperture, and darker skies all increase SNR.

"Depth" in this document means SNR on the faint, extended structure of a target (nebulousity, galaxy arms, tidal streams).

The key modelling decision is to measure SNR **per unit area of sky** - per fixed patch of the sky as seen from Earth (for example, per square arcsecond) - rather than **per pixel**.

Why per area and not per pixel? Because per-pixel SNR depends on how finely you sample the sky, which is an arbitrary choice of optics and sensor, not a property of how much real data you collected:

- A long focal length spreads a patch of sky across many pixels. Each pixel receives fewer photons, so per-pixel SNR drops - but you have more pixels.
- A short focal length crams the same patch onto fewer pixels. Each pixel gets more photons, so per-pixel SNR rises - but you have fewer pixels.

In both cases the *total* light collected from that patch of sky is the same. If you measure SNR over a fixed area of sky (summing or averaging across whatever pixels happen to cover it), the arbitrary sampling choice cancels out, and you are left with a number that reflects genuine data depth. This is exactly how experienced imagers think when they say a fast, small scope and a slow, large scope "go equally deep per unit time" - they are comparing area-normalised SNR.

This single choice is what makes the whole comparison fair and is the foundation of everything below.

3. The physics, from photons to a comparable number

Consider one fixed patch of sky containing part of the target. Over an exposure of duration t , the telescope collects photons from two sources: the target and the sky background.

Signal

The number of target photons detected from that patch is proportional to:

$$S \propto A \cdot \Sigma_{\text{target}} \cdot QE \cdot T \cdot \Delta\lambda \cdot t$$

where:

- **A** = collecting area of the aperture ($\propto$ the square of the aperture diameter D)
- **Σ_{target}** = surface brightness of the target in that patch (a property of the object, identical for everyone imaging it)
- **QE** = quantum efficiency of the sensor (fraction of photons detected)
- **T** = transmission of the filter (fraction of light passed)
- **$\Delta\lambda$** = bandwidth - the width of the spectral window the filter admits
- **t** = exposure time

Noise

Deep-sky imaging from any real sky is almost always **background-limited**: the dominant source of noise is the random (Poisson) fluctuation in the sky-glow photons, not the camera's read noise or dark current. For a Poisson process the noise equals the square root of the number of events, so the noise from the sky background in that patch is:

$$N \propto \sqrt{(A \cdot \Sigma_{\text{sky}} \cdot QE \cdot T \cdot \Delta\lambda \cdot t)}$$

where **Σ_{sky}** is the sky surface brightness (how bright the background glow is).

Signal-to-noise ratio

Dividing signal by noise:

$$\text{SNR} = S / N \propto \Sigma_{\text{target}} \cdot \sqrt{(A \cdot QE \cdot T \cdot \Delta\lambda \cdot t / \Sigma_{\text{sky}})}$$

Two things fall out of this immediately:

1. **The target's surface brightness Σ_{target} is a constant** for everyone photographing the same object. When we compare two images of the same target, it cancels. That is why the target itself never appears in any of the extension's formulas - and why the tool works without knowing what the object is.

2. **SNR grows as the square root of (aperture area $\times$ throughput $\times$ time $\div$ sky).**

Because area $\propto D^2$, the square root of area is $\propto D$, the aperture diameter. This

is the origin of every term in the engine.

We define **depth** as the part of SNR that depends on the gear, the sky, and the time (everything except the cancelling target term):

$$\text{depth} \propto D \cdot \sqrt{(QE \cdot T \cdot \Delta\lambda \cdot t / \Sigma_{\text{sky}})} \quad (\text{broadband})$$

4. The depth formulas

It is convenient to separate the *rate* at which depth accumulates from the time itself. Since depth grows as $\sqrt{t}$, we write:

$$\text{depth} = \text{coeff} \cdot \sqrt{t}$$

where **coeff** is the per-rig "depth rate" - how fast a given telescope, sensor, filter and sky build up signal-to-noise. It does not depend on time.

The two regimes (broadband and emission line) differ in how bandwidth enters.

Broadband (luminance, R, G, B)

For broadband light, both the target and the sky span the filter's band, so a wider band collects proportionally more of both. The signal grows with $\Delta\lambda$ and the sky noise grows with $\sqrt{\Delta\lambda}$, leaving a net $\sqrt{\Delta\lambda}$ benefit:

$$\text{coeff_broadband} = D \cdot \sqrt{(QE \cdot T \cdot \Delta\lambda / \Sigma_{\text{sky}})}$$

A wider broadband filter therefore goes deeper per unit time (more total photons), which is why luminance - the widest band - dominates broadband depth.

Emission line (H α , OIII, SII)

For a narrowband emission line, essentially all of the target's light sits inside a very thin spectral line. As long as the filter is at least as wide as the line, narrowing the filter further does **not** cut the target signal - but it *does* cut the sky background that leaks through. So for a line, the signal is independent of bandwidth, while the sky noise still grows with $\sqrt{\Delta\lambda}$:

$$\text{coeff_line} = D \cdot \sqrt{(QE \cdot T / (\Sigma_{\text{sky}} \cdot \Delta\lambda))}$$

Here a *narrower* filter goes deeper, because it rejects more skyglow and moonlight while preserving the line. This is the whole reason 3 nm narrowband filters outperform 7 nm filters under bright skies.

Sky flux

The sky surface brightness Σ_{sky} is derived from the sky-quality measure SQM (magnitudes per square arcsecond) using the standard magnitude-to-flux relation:

$$\Sigma_{\text{sky}} \propto 10^{(-0.4 \cdot \text{SQM})}$$

A higher SQM means a darker sky and exponentially less background flux. Each 1.0 mag/arcsec² of darker sky is about 2.5× less sky flux.

Matching an image

To find how long *your* rig needs to reach the same depth as an image, set your depth equal to the image's depth and solve for time:

$$\begin{aligned} \text{depth_image} &= \text{coeff_you} \cdot \sqrt{\text{your_time}} \\ \square \quad \text{your_time} &= (\text{depth_image} / \text{coeff_you})^2 \end{aligned}$$

5. The translator: converting one rig's time to another's

Combining the two relations above, the time your rig needs to match an image is:

$$\text{your_time} = \text{their_time} \cdot (\text{coeff_them} / \text{coeff_you})^2$$

Expanding the coefficient ratio for broadband gives the intuitive, fully explicit translator formula the extension shows on hover:

$$\begin{aligned} \text{your_time} = \text{their_time} & \\ & \cdot (D_{\text{them}} / D_{\text{you}})^2 && \leftarrow \text{aperture: collecting-area ratio} \\ & \cdot (QE \cdot T \cdot \Delta\lambda)_{\text{them}} / (QE \cdot T \cdot \Delta\lambda)_{\text{you}} && \leftarrow \text{sensor + filter efficiency} \\ & \cdot (\Sigma_{\text{sky, you}} / \Sigma_{\text{sky, them}}) && \leftarrow \text{your sky vs. their sky} \end{aligned}$$

For an emission line the bandwidth term flips (it divides rather than multiplies, per the line coefficient), and a Bayer-area term is added for colour sensors (Section 8).

Each factor has a plain-language meaning:

- **Aperture ratio (squared):** a scope with half the aperture needs four times the exposure, because collecting area scales with the square of diameter.
- **Sensor + filter efficiency:** a more sensitive sensor or higher-throughput filter shortens the required time proportionally.
- **Sky ratio:** imaging from a brighter sky multiplies the time needed; from a darker sky, it divides it.

Crucially, the **target's brightness cancels** (it is the same object), so the translator never needs to know what the object is.

6. Why focal length and pixel size do not appear

A frequent and reasonable question: *surely pixel size, focal length, and f-ratio matter?* For framing and resolution, yes. For per-area depth, no - and this is a direct consequence of the per-area choice in Section 2.

- **Focal length** changes the image scale: it determines how much sky lands on each pixel, i.e. how big the target appears. It does not change how many target photons the aperture collects from a given patch of sky.
- **Pixel size** changes how that fixed bucket of photons is divided among pixels. Smaller pixels split it into more, fainter pixels; larger pixels into fewer, brighter ones. Summed over a fixed area of sky, the total is unchanged.
- **F-ratio** (focal length $\div$ aperture) is famous for setting per-pixel exposure speed - the "f-ratio myth" debate in astrophotography. Per pixel, a faster f-ratio is brighter. But per unit area of sky, only the aperture (total light grasp) and throughput matter. Two scopes of equal aperture reach the same per-area depth in the same time regardless of f-ratio; the faster one simply packs that depth into a wider field at coarser resolution.

So the reference rig and the engine deliberately ignore focal length and pixel size for the depth calculation. They are not thrown away, though: the extension separately compares your pixel scale (computed as $206.265 \times \text{pixel_}\mu\text{m} \div \text{focal_mm}$,

in arcseconds per pixel) to the image's, and reports whether your setup would sample the target more finely or more coarsely. That is a statement about *resolution*, kept distinct from *depth*.

7. Bands, not filters

Real images are built from several channels. The extension reasons in terms of **bands** rather than individual filter brands, and never mixes the two physical regimes into one number.

- **Broadband, mono camera:** each of L, R, G, B is captured separately. The tool reports the time per channel and a broadband subtotal (their sum, since they are shot one after another).
- **Broadband, colour (OSC) camera:** R, G, B are captured simultaneously in one exposure through the Bayer matrix, so broadband is reported as a single combined number.
- **Narrowband:** each emission line (H α , OIII, SII) is treated on its own, with its own depth and its own required time, because the lines are physically independent and a filter that is great for one may be useless for another.

When an image combines broadband and narrowband data, the two are ****never blended into a single depth scalar**** - adding an SNR figure from a 300 nm luminance band to one from a 3 nm OIII line would be physically meaningless. They are shown as separate groups. (A total *clock time* across all bands is still meaningful and is shown as the headline - see Section 11 - because time is additive even when depth is not.)

To combine the broadband channels into one broadband depth, their individual depths are added in quadrature (root-sum-square), which is the correct way to combine independent signal-to-noise contributions:

$$\text{depth_broadband_total} = \sqrt{(\text{depth_L}^2 + \text{depth_R}^2 + \text{depth_G}^2 + \text{depth_B}^2)}$$

8. Colour sensors and the Bayer matrix

A one-shot-colour (OSC) sensor places a red, green, or blue micro-filter over each pixel in a repeating mosaic (the Bayer matrix: 25% red, 50% green, 25% blue). This has a large effect on narrowband performance that the model accounts for explicitly.

- For an **emission line** on a colour sensor, only the pixels whose colour matches the line actually detect it. H α and SII (deep red) are seen mainly by the 25% of pixels that are red; OIII (blue-green) by roughly the 50% that are green (and some blue). The model multiplies the native pixel QE by this ****Bayer area fraction**** (0.25 for H α /SII, 0.50 for OIII), because only that fraction of the sensor area is collecting the line. This is why a colour camera with a dual-band filter is dramatically slower on narrowband than a mono camera with dedicated filters - often by 4 $\times$ or more - and the tool will flag truly impractical cases.
- For **broadband** on a colour sensor, the Bayer loss is already baked into the effective broadband quantum efficiency (qe_lum) stored for that camera, so no additional area factor is applied - applying it twice would double-count.

For mono sensors the Bayer fraction is 1.0 throughout (every pixel sees every band, through whichever filter is mounted).

9. Sky brightness: SQM, Bortle, and the Moon

Sky brightness enters every formula through Σ_{sky} , and the extension determines it in order of preference:

1. **Measured SQM** if the image lists one - the most accurate input, used directly.
2. **Bortle class** if only that is given - converted to a representative SQM via a standard lookup table (for example Bortle 4 $\approx$ 21.2, Bortle 6 $\approx$ 19.9, Bortle 8 $\approx$ 18.4 mag/arcsec²). This is an approximation, and the confidence score is reduced accordingly (Section 13).

3. **A dark-sky default** (SQM $\approx$ 21.0) if nothing is listed, clearly noted.

Narrowband and skyglow

Narrowband filters reject the vast majority of light pollution and moonlight, because city lighting and the Moon emit little at the exact wavelengths of the target lines. A bright broadband sky therefore does **not** translate into a bright narrowband sky. The engine enforces a **narrowband sky floor** (treating the narrowband sky as no worse than roughly SQM 21.3 regardless of Bortle), so it will not wrongly punish narrowband imaging from a city. This matches the lived experience that narrowband works well even under heavy light pollution.

Moonlight

The Moon brightens the **broadband** sky only. The extension models moonlight as an additive reduction in the effective SQM that grows non-linearly toward full Moon (using the square of the illuminated fraction as a coarse proxy, since moonlit sky brightness rises steeply near full). It is applied to broadband bands only; narrowband is left on its floor. The Moon contribution is **off by default** and exposed as a checkbox, because most serious deep-sky work is planned around the Moon; ticking it factors in the image's listed lunar phase.

10. The reference rig (the universal yardstick)

To make any two images comparable regardless of who shot them, the extension expresses every image's depth as the equivalent time on a single fixed **reference rig** - the "reference-hours" figure. The reference is:

- **203 mm (8") aperture** - a common, mid-range deep-sky aperture.
- **An ASI2600MM-class monochrome sensor**, with realistic per-band quantum efficiencies.
- **7 nm narrowband filters** and standard broadband filters at typical transmissions.
- **Bortle 4 sky (SQM $\approx$ 21.2), no Moon.**

Because focal length and pixel size do not affect per-area depth (Section 6), the reference does not need to specify them. The reference is a tunable constant: it is the yardstick, not a claim that everyone should own this rig. Its only purpose is to give every image a number on the same scale, so that "31 reference-hours" means the same depth whether the image was shot on a 60 mm refractor or a metre-class telescope.

11. The headline number: total shooting time

The large headline figure is the **total** reference-hours across all bands - the whole time the reference scope would spend shooting to reach this image's depth in every band it contains: the combined broadband group plus each narrowband line, summed.

This is deliberately a sum of clock time, not a single depth. As noted in Section 7, depths across different band kinds cannot be added, but the *time* spent acquiring them can - and "how long in total" is the most intuitive headline for "how big a project is this." Beneath the headline, a breakdown line splits it into its broadband and narrowband portions (for example "6 min broadband + 31 h 30 m narrowband"), so the dominant cost is always visible.

The per-rig table below the headline mirrors this: it lists each band's time on your gear, a broadband subtotal, a narrowband subtotal, and a grand total - indented detail rows under flush subtotals, so the hierarchy reads at a glance.

12. Multi-line filters: dual-band vs. filter sets

A subtlety that the extension gets right, and which is easy to get wrong: narrowband total time depends on *how* the lines are captured.

- **A true dual- or tri-band filter** (for example an Optolong L-eXtreme, which passes H α and OIII through one piece of glass) captures all of its lines ****in a single exposure, simultaneously.**** The total narrowband time is therefore the

time of the **slowest** line it covers - the others come "for free" in the same sub-exposures. The engine groups the lines served by one such filter and takes the maximum.

- **A filter set or bundle** (for example an SHO set of separate 7 nm H α , OIII, and SII filters) is **three separate filters, shot one at a time, sequentially**.

Their times **add up**. The engine sums them.

Confusing the two understates a set's real cost dramatically. For example, with a sequential SHO set, 8 hours of H α plus 25 hours of OIII is a 33-hour project - not a 25-hour one. The extension distinguishes the cases with a set flag, inferred from the filter being a recognised set/bundle, or carrying three or more emission lines (no single interference filter usefully passes H α , OIII, and SII at once), or having "set"/"bundle" in its name. Dual/tri-band filters default to the simultaneous (max) treatment; sets to the sequential (sum) treatment.

13. The confidence score

Every estimate is only as good as its inputs, so the panel shows a confidence badge with a hover breakdown of four checks:

1. **Image camera QE** - was the sensor matched to a known quantum efficiency, or estimated?
2. **Sky brightness** - a measured SQM is fully solid; a Bortle class is a good approximation (partial credit); nothing listed is a gap.
3. **Image filters** - were the bandwidths and transmissions identified, or assumed from standard specs?
4. **Your rig specs** - is your own gear fully entered?

Each check contributes a score: 1.0 for a solid input, 0.75 for a reasonable approximation (such as sky-from-Bortle), 0.0 for missing. The overall rating is the total score divided by the number of checks:

- $\geq 0.95 \rightarrow$ **High**
- $\geq 0.80 \rightarrow$ **Medium-High**
- $\geq 0.50 \rightarrow$ **Medium**

- **otherwise → Low**

So an image with everything known except a measured SQM (Bortle given instead) lands at **Medium-High**; with a real SQM it reaches **High**; with no sky data at all it drops to **Medium**. The badge is colour-coded and the hover lists each check with a ✓ (solid), ≈ (approximate), or ✕ (missing) marker, so the user always knows how much to trust the number.

14. A fully worked example

Suppose an image was shot on a 203 mm scope with a mono ASI2600MM-class sensor under a Bortle 4 sky, with 12 h of luminance and 6 h of H α through a 7 nm filter, and you want to know the time on the same class of rig.

For luminance (broadband), the coefficient is $\text{coeff} = D \cdot \sqrt{(QE \cdot T \cdot \Delta\lambda / \Sigma_{\text{sky}})}$.

Because your rig matches the reference closely, your luminance coefficient is near the reference's, so your luminance time comes out near the original 12 h, and R/G/B each near their listed times. Summed, that is the broadband subtotal.

For H α (a line), $\text{coeff} = D \cdot \sqrt{(QE \cdot T / (\Sigma_{\text{sky}} \cdot \Delta\lambda))}$. With the same aperture, sensor, 7 nm filter and sky, your H α time again lands near the original 6 h.

The headline reference-hours is the sum of the combined broadband group and the H α line - the whole shooting project on the yardstick rig. Because every input is known except the sky is given as Bortle rather than a measured SQM, the confidence reads **Medium-High**.

Now change one thing: image the *same* target from your back garden at Bortle 7 instead of Bortle 4. The sky ratio term $\Sigma_{\text{sky, you}} / \Sigma_{\text{sky, them}}$ rises sharply for the broadband channels (skyglow is far worse), multiplying your broadband time several-fold - while the H α time barely moves, because the narrowband sky floor shields it from the light pollution. The panel would show the broadband ballooning and the narrowband almost unchanged, exactly as a real imager would experience.

15. Assumptions and limitations

The model is a physically-grounded estimate, not a laboratory measurement. Its main simplifying assumptions:

- **Background-limited imaging.** It assumes sky-background noise dominates read noise and dark current. This holds for typical deep-sky sub-exposures from a real sky, but breaks down for very short subs, extremely narrow filters under truly dark skies, or sensors with unusually high read noise. Two cameras with identical QE but different read noise are treated as equal.
 - **Read noise and dark current are not modelled.** Only QE, transmission, bandwidth, aperture, sky, and time enter.
 - **Filter passes the whole line.** For emission lines it assumes the filter is at least as wide as the target line, so narrowing rejects only sky, not signal.
 - **Quantum efficiencies are catalogue/peak values,** from a built-in table or estimated when unknown; AstroBin does not publish QE.
 - **Filter transmission and bandwidth are sometimes assumed.** AstroBin publishes bandwidth for some filters but never transmission, so standard per-type values are used and disclosed in the notes.
 - **Surface brightness is uniform within a patch.** The per-area treatment is an average; it does not model how SNR varies across bright cores vs. faint outskirts.
 - **It estimates depth, not quality.** Framing, focus, guiding, seeing, processing skill, and artistic merit are entirely outside its scope.
 - **Outputs are only as good as the metadata** each image's author entered on AstroBin. Missing or wrong gear/sky data propagates into the estimate, which is why the confidence score exists.
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16. Where the numbers come from (data sources)

- **The image's acquisition card on the page** is read directly from the rendered AstroBin image viewer: integration time per channel, the equipment list, and the sky/pixel-scale details in the astrometry header. Every selector has a text-based fallback so a markup change degrades gracefully rather than breaking.

- **AstroBin's own public equipment API** is queried (the same data the website itself loads) to fill in aperture and focal length for telescopes, and bandwidth and sensor details for cameras and filters, when a piece of gear is not in the built-in table. Results are cached locally so the same item is never requested twice.

- **A built-in equipment table** supplies the specifications AstroBin does not publish - principally sensor quantum efficiencies and filter transmissions - for common cameras, scopes, and filters, plus the user's own gear. It is editable, and any gear you enter is remembered privately in your browser.

No scraping of AstroBin's protected data is performed; only the public API and the already-rendered page content are used.

17. Software architecture

The extension is a Manifest V3 Chrome extension with no build step - plain JavaScript, injected as content scripts on AstroBin image pages.

- `equipment.js` - the data layer: the camera/scope/filter tables, the Bortle→SQM table, the reference rig, and the physical constants (Bayer fractions, moon model, narrowband sky floor, practical-time limit). Exposes `window.ADT_DATA`.
- `engine.js` - the calculation engine: the per-area depth model, the coefficient functions, the translator, the reference-hours and subtotal computations, and the confidence score. Pure functions, independently testable in Node. Exposes `window.ADT_ENGINE`.
- `parser.js` - reads the rendered AstroBin page into an engine-ready image object (channels, equipment, sky), with robust fallbacks. Exposes `window.ADT_PARSER`.
- `content.js` - orchestration and UI: builds the panel, the inline gear settings (type-ahead search backed by the AstroBin API, raw-spec entry, private custom gear, multi-filter chips), draws the results table, and watches the single-page app for navigation so the panel re-renders on each image.

- `panel.css` - styling that mirrors AstroBin's dark theme.
- `background.js` - a minimal service worker that exists only to relay the optional, off-by-default analytics events (Section 18).

The scripts load in dependency order (`data` → `engine` → `parser` → `content`) and communicate through the three `window.ADT_*` globals. Editing any file requires reloading the unpacked extension and refreshing the AstroBin tab to re-inject the updated content script.

18. Privacy and analytics

The extension is built to run entirely in the browser. It reads only the AstroBin page you are viewing and AstroBin's public equipment API; it accesses no other sites and reads no browsing history. Your gear settings and the cached equipment specs are stored locally in Chrome's `storage.local` and never leave your device. It requires no account or login.

There is an **optional, off-by-default** anonymous analytics toggle. Only if you explicitly tick it (and only if the developer has configured a collection endpoint, which ships empty) does it send anonymous events - a random install ID, which gear and settings are used, and error messages - to help improve the tool. It never includes your name, your AstroBin account, image URLs, or any personal data, and can be turned off at any time. This honest opt-in design was a deliberate choice over any hidden or pre-ticked collection, both because it is the right thing to do and because deceptive data collection is prohibited by the Chrome Web Store and would destroy trust with a technical audience.

19. Glossary of symbols

- **D** - aperture diameter (mm). Collecting area $A \propto D^2$.
- **QE** - quantum efficiency; fraction of incident photons detected (0-1).
- **T** - filter transmission; fraction of light passed (0-1).
- **$\Delta\lambda$** - filter bandwidth (nm); the width of the spectral window.

